Question "Orange"

1.1)
First of all, we use the Gauss's law for a single plate to obtain the electric field,

$E = \frac{\sigma}{\varepsilon_0}$. (0.2)

The density of surface charge for a plate with charge, Q and area, A is

$\sigma = \frac{Q}{A}$. (0.2)

Note that the electric field is resulted by two equivalent parallel plates. Hence the contribution of each plate to the electric field is $\frac{1}{2}E$. Force is defined by the electric filed times the charge, then we have

Force = $\frac{1}{2}EQ = \frac{Q^2}{2\varepsilon_0 A}$ (0.2)+ (0.2) (The ½ coefficient + the final result)

1.2)
The Hook's law for a spring is

$F_m = -k\,x$. (0.2)

In 1.2 we derived the electric force between two plates is

$F_e = \frac{Q^2}{2\varepsilon_0 A}$.

The system is stable. The equilibrium condition yields

$F_m = F_e$, (0.2)

$\Rightarrow \qquad x = \frac{Q^2}{2\varepsilon_0 A k}$ (0.2)

1.3)
The electric field is constant thus the potential difference, V is given by

$V = E(d - x)$ (0.2)

(Other reasonable approaches are acceptable. For example one may use the definition of capacity to obtain V.)

By substituting the electric field obtained from previous section to the above equation, we get, $V = \frac{Qd}{\varepsilon_0 A}\left(1 - \frac{Q^2}{2\varepsilon_0 A k d}\right)$ (0.2)

1.4)
C is defined by the ratio of charge to potential difference, then

$C = \frac{Q}{V}$. (0.1)

Using the answer to 1.3, we get $\frac{C}{C_0}=\left(1-\frac{Q^2}{2\varepsilon_0 A\,k\,d}\right)^{-1}$ (0.2)

1.5)
Note that we have both the mechanical energy due to the spring

$U_m=\frac{1}{2}kx^2$, (0.2)

and the electrical energy stored in the capacitor.

$U_E=\frac{Q^2}{2C}$. (0.2)

Therefore the total energy stored in the system is

$U=\frac{Q^2 d}{2\varepsilon_0 A}\left(1-\frac{Q^2}{4\varepsilon_0 A\,k\,d}\right)$ (0.2)

2.1)
For the given value of x, the amount of charge on each capacitor is

$Q_1=V\,C_1=\frac{\varepsilon_0 AV}{d-x}$, (0.2)

$Q_2=V\,C_2=\frac{\varepsilon_0 AV}{d+x}$. (0.2)

2.2)
Note that we have two capacitors. By using the answer to 1.1 for each capacitor, we get

$F_1=\frac{Q_1^{\,2}}{2\varepsilon_0 A}$,

$F_2=\frac{Q_2^{\,2}}{2\varepsilon_0 A}$.

As these two forces are in the opposite directions, the net electric force is

$F_E=F_1-F_2$, (0.2) $\Rightarrow$ $F_E=\frac{\varepsilon_0 AV^2}{2}\left(\frac{1}{(d-x)^2}-\frac{1}{(d+x)^2}\right)$ (0.2)

2.3)
Ignoring terms of order x^2 in the answer to 2.2., we get

$F_E=\frac{2\varepsilon_0 AV^2}{d^3}x$ (0.2)

2.4)
There are two springs placed in series with the same spring constant, k, then the mechanical force is

$F_m = -2kx$. (The coefficient (2) has (0.2))

Combining this result with the answer to 2.4 and noticing that these two forces are in the opposite directions, we get

$F = F_m + F_E$, $\Rightarrow$ $F = -2\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right)x$, (Opposite signs of the two forces have (0.3))

$\Rightarrow \quad k_{eff} = 2\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right)$ (0.2)

2.5)

By using the Newtown's second law,

$F = ma$ (0.2)

and the answer to 2.4, we get

$a = -\frac{2}{m}\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right)x$ (0.2)

3.1)

Starting with Kirchhoff's laws, for two electrical circuits, we have

$$\begin{cases} \frac{Q_S}{C_S} + V - \frac{Q_2}{C_2} = 0 \\ -\frac{Q_S}{C_S} + V - \frac{Q_1}{C_1} = 0 \\ Q_2 - Q_1 + Q_S = 0 \end{cases}$$

(Each has (0.3), Note: the sings may depend on the specific choice made)

Noting that $V_s = \frac{Q_S}{C_S}$ one obtains

$$\Rightarrow V_S = V \frac{\frac{2\varepsilon_0 A x}{d^2 - x^2}}{C_S + \frac{2\varepsilon_0 A d}{d^2 - x^2}}.$$ ((0.4) + (0.2): (0.4) for solving the above equations and (0.2) for final result)

Note: Students may simplify the above relation using the approximation $d^2 >> x^2$. It does not matter in this section.

3.2)

Ignoring terms of order x^2 in the answer to 3.1., we get

$$V_S = V\frac{2\varepsilon_0 A\,x}{d^2 C_S + 2\varepsilon_0 A\,d}\ . \qquad (0.2)$$

4.1)

The ratio of the electrical force to the mechanical (spring) force is

$$\frac{F_E}{F_m} = \frac{\varepsilon_0 A V^2}{k\,d^3}\ ,$$

Putting the numerical values:

$$\frac{F_E}{F_m} = 7.6\times10^{-9}\ .$$ ((0.2) + (0.2) + (0.2): (0.2) for order of magnitude, (0.2) for two significant digits and (0.2) for correct answer (7.6 or 7.5)).

As it is clear from this result, we can ignore the electrical forces compared to the electric force.

4.2)

As seen in the previous section, one may assume that the only force acting on the moving plate is due to springs:

$F = 2k\,x$. (The concept of equilibrium (0.2))

Hence in mechanical equilibrium, the displacement of the moving plate is

$$x = \frac{ma}{2k}\ .$$

The maximum displacement is twice this amount, like the mass spring system in a gravitational force field, when the mass is let to fall.

$x_{\max} = 2\,x$ (0.2)

$$x_{\max} = \frac{m\,a}{k} \qquad (0.2)$$

4.3)

At the acceleration

$a = g$, (0.2)

The maximum displacement is

$$x_{\max} = \frac{m\,g}{k}\ .$$

Moreover, from the result obtained in 3.2, we have

$$V_S = V\frac{2\varepsilon_0 A\, x_{\max}}{d^2 C_S + 2\varepsilon_0 A\, d}$$

This should be the same value given in the problem, $0.15\,\mathrm{V}$.

$$\Rightarrow \qquad C_S = \frac{2\varepsilon_0 A}{d}\left(\frac{V\, x_{\max}}{V_S\, d} - 1\right) \qquad (0.2)$$

$$\Rightarrow \qquad C_S = 8.0\times 10^{-11}\,\mathrm{F} \qquad (0.2)$$

4.4)

Let ℓ be the distance between the driver's head and the steering wheel. It can be estimated to be about

$\ell = 0.4\,m - 1\,m$. (0.2)

Just at the time the acceleration begins, the relative velocity of the driver's head with respect to the automobile is zero.

$\Delta v(t=0) = 0$, (0.2)

then

$$\ell = \frac{1}{2} g\, t_1^{\,2} \qquad \Rightarrow \qquad t_1 = \sqrt{\frac{2\ell}{g}} \qquad (0.2)$$

$t_1 = 0.3 - 0.5\ s$ (0.2)

4.5)

The time t_2 is half of period of the harmonic oscillator, hence

$$t_2 = \frac{T}{2}, \qquad (0.3)$$

The period of harmonic oscillator is simply given by

$$T = 2\pi\sqrt{\frac{m}{2k}}\ , \qquad (0.2)$$

therefore,

$t_2 = 0.013\ s$. (0.2)

As $t_1 > t_2$, the airbag activates in time. (0.2)